

Real-Time Stock Market Movement Prediction Using Recurrent Neural Network Architectures with Sentiment Analysis

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Abstract

Predicting the directional movement of financial markets is a fundamental challenge in quantitative finance, requiring models capable of capturing complex temporal dependencies embedded in noisy, non-stationary price time-series. This paper presents a comparative study of four recurrent neural network (RNN) architectures—Simple RNN, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM)—applied to binary next-day price direction classification for Apple Inc. (AAPL) and a basket of four major technology equities. Each model is trained on a rich feature set comprising 20 technical indicators derived from five years of OHLCV data (2020–2024), augmented with daily sentiment scores extracted from financial news RSS feeds using the VADER lexicon. A sliding window of 30 trading days constructs the input sequences. Models are evaluated on accuracy, F1-score, and root mean square error (RMSE). Experimental results demonstrate that the BiLSTM model achieves superior predictive performance by leveraging bidirectional temporal context, while GRU offers a competitive accuracy-to-complexity trade-off. A lightweight Flask-based inference API is deployed to serve real-time predictions given a ticker symbol and an optional news headline, validating the practical applicability of the proposed framework.

Index Terms—stock market prediction, recurrent neural networks, LSTM, GRU, bidirectional LSTM, sentiment analysis, VADER, technical indicators, time-series classification, deep learning.

I. INTRODUCTION

Financial markets generate vast volumes of heterogeneous data every trading session, including price quotations, order flows, macroeconomic releases, and an ever-expanding body of news and social media commentary. Extracting actionable directional signals from this noisy, non-stationary stream is one of the most studied and commercially relevant problems in computational finance [1].

Classical approaches such as ARIMA and linear regression models assume stationarity and fail to capture the long-range dependencies and non-linear interactions inherent in financial time-series. Deep learning—particularly recurrent architectures—has emerged as a compelling alternative because such models can autonomously learn multi-scale temporal abstractions without explicit feature engineering [2].

Despite the proliferation of individual model proposals, rigorous side-by-side comparisons of RNN variants under identical data preprocessing, feature engineering, and evaluation protocols remain rare. Moreover, jointly modelling price-derived technical signals and news sentiment—two information channels known to move markets—is seldom studied in a unified reproducible framework.

A. Contributions

This paper makes the following contributions:

- A unified data pipeline ingesting five years of Yahoo Finance price data for five technology tickers, computing 20 technical indicators, and merging daily sentiment aggregates derived from Reuters RSS feeds.
- Systematic implementation and hyperparameter-controlled training of four recurrent architectures (RNN, LSTM, GRU, BiLSTM) under identical conditions, enabling fair comparison.
- Quantitative evaluation using accuracy, F1-score, and RMSE, with class-weight balancing to address the moderate label imbalance observed in the dataset.
- Deployment of the best-performing model as a Flask REST API that accepts a ticker symbol and an optional live news headline, enabling real-time directional inference.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset and feature engineering methodology. Section IV details each model architecture. Section V presents experimental results and discussion. Section VI describes the deployment. Section VII concludes the paper.

II. RELATED WORK

The application of deep learning to stock market prediction has a rich literature. Hochreiter and Schmidhuber [3] introduced the LSTM cell with gating mechanisms that mitigate the vanishing gradient problem endemic to simple RNNs, enabling models to retain information over hundreds of time steps. Cho et al. [4] proposed the GRU as a parameter-efficient simplification of LSTM that achieves comparable performance on sequential modelling tasks.

Ding et al. [5] demonstrated that fusing news event representations with price features improves directional forecasting accuracy relative to price-only baselines. Fischer and Krauss [6] conducted a large-scale evaluation of LSTM on S&P 500 constituents and found statistically significant predictive ability over a random walk. Schuster and Paliwal [7] introduced bidirectional RNNs, later applied to financial series by several authors who showed that accessing future context within a window improves classification accuracy.

Hutto and Gilbert [8] developed the VADER lexicon, a rule-based sentiment analyser tuned for social media text that has been widely adopted for rapid financial news sentiment scoring due to its zero-shot performance and computational efficiency. Our work synthesises these threads by implementing all four architectures under a common pipeline and evaluating them on sentiment-augmented technical feature sequences.

III. DATASET AND FEATURE ENGINEERING

A. Data Sources

Stock price data for five NASDAQ-listed technology equities—Apple (AAPL), Alphabet (GOOGL), Microsoft (MSFT), Amazon (AMZN), and Tesla (TSLA)—was downloaded from Yahoo Finance for the period 2020-01-01 to 2024-12-31 using the yfinance API. Daily OHLCV (Open, High, Low, Close, Volume) records were retrieved, yielding approximately 1,260 trading days per ticker.

Financial news was collected from the Reuters RSS feed. For each article, the headline and summary were concatenated and passed through the VADER SentimentIntensityAnalyzer to obtain a compound score in the range $[-1, 1]$. Scores were mapped to discrete labels: Positive (compound ≥ 0.05), Negative (compound ≤ -0.05), and Neutral otherwise.

B. Technical Indicators

Twenty technical features were computed from the AAPL OHLCV series:

Category	Indicator	Parameters
Trend	Simple Moving Average	MA(5), MA(20), MA(50)
Trend	Exponential Moving Average	EMA(12), EMA(26)
Momentum	MACD + Signal Line	EMA(12)–EMA(26), EMA(9) of MACD
Momentum	Relative Strength Index	Window = 14
Volatility	Bollinger Bands (Width, Upper, Lower)	Window = 20, $\sigma = 2$
Volatility	Historical Volatility	Window = 5, 20 days
Volume	Volume Ratio	Volume / MA(20) Volume
Sentiment	Daily Mean Sentiment Score	VADER compound, aggregated
Sentiment	Positive / Negative Ratios	Fraction of articles per day

Table I: Feature set used as model input.

C. Sequence Construction and Label Definition

A sliding window of length $L = 30$ trading days was applied to construct input tensors $X \in \mathbb{R}^{30 \times F}$ where F is the number of features. The binary target label $y = 1$ if $\text{Close}_{t+1} > \text{Close}_t$ (price goes up) and $y = 0$ otherwise. The dataset was split into training (70%), validation (15%), and test (15%) sets in chronological order to prevent look-ahead bias. Feature values were scaled to $[0, 1]$ using MinMaxScaler fitted exclusively on the training partition.

D. Class Imbalance

The target distribution shows a slight up-day bias (~52% positive) reflecting the secular bull trend in technology equities over the study period. To mitigate this, class weights inversely proportional to class frequency were applied during model training.

IV. MODEL ARCHITECTURES

A. Simple Recurrent Neural Network (RNN) — Khushbakht Adnan (F230005)

The Simple RNN processes the input sequence through three stacked recurrent layers (128 → 64 → 32 hidden units) with tanh activations. Batch normalisation is applied after each recurrent layer, and Dropout(0.2) is inserted for regularisation. The final representation is passed through a Dense(32, ReLU) layer followed by a sigmoid output neuron for binary classification:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$

A known limitation of simple RNNs is the vanishing gradient problem, which can impede the learning of dependencies spanning more than a few time steps. This is relevant in our 30-step windows where market momentum signals may persist over extended periods.

B. Long Short-Term Memory (LSTM) — Urwa Tehseen (F230008)

The LSTM [3] extends the simple RNN with three gating mechanisms—input gate i_t , forget gate f_t , and output gate o_t —and a separate cell state c_t that carries long-range information with minimal gradient attenuation:

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{c}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad h_t = o_t \odot \tanh(c_t) \end{aligned}$$

The architecture mirrors the RNN stack (128 → 64 → 32 units) with identical regularisation and classification head components.

C. Gated Recurrent Unit (GRU) — Noor ul Huda (F230035)

The GRU [4] consolidates the LSTM's four gates into two—a reset gate r_t and an update gate z_t —reducing parameter count while retaining the ability to model long-range dependencies:

$$\begin{aligned} z_t &= \sigma(W_z \cdot [h_{t-1}, x_t]), \quad r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \\ \tilde{h}_t &= \tanh(W \cdot [r_t \odot h_{t-1}, x_t]), \quad h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \end{aligned}$$

The GRU stack follows the same 128 → 64 → 32 unit progression and regularisation scheme as LSTM, allowing a controlled parameter-count comparison.

D. Bidirectional LSTM (BiLSTM) — Zeenat Ijaz (F230054)

The BiLSTM [7] wraps each LSTM cell in a bidirectional layer that processes the input sequence in both forward and backward directions, concatenating both hidden states:

$$h_t = [h_t^{\rightarrow}; h_t^{\leftarrow}]$$

Within a 30-day window, the backward pass allows the model to contextualise early sequence positions with information from later time steps, potentially capturing reversal and consolidation patterns. The BiLSTM uses 64 → 32 → 16 units per direction (equivalent parameter budget to the unidirectional models after concatenation), with the same Dropout and Dense head design.

E. Training Configuration

Hyperparameter	Value
Optimiser	Adam ($\text{lr} = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Loss function	Binary cross-entropy with class weights
Batch size	16
Max epochs	100 (early stopping applied)
Early stopping patience	20 epochs on validation loss
LR reduction on plateau	Factor 0.5, patience 10, $\text{min_lr} = 1\text{e-}6$
Sequence length (L)	30 trading days
Dropout rate	0.2 after each recurrent block
Batch normalisation	After each recurrent layer

Table II: Common training hyperparameters for all models.

V. EXPERIMENTS AND RESULTS

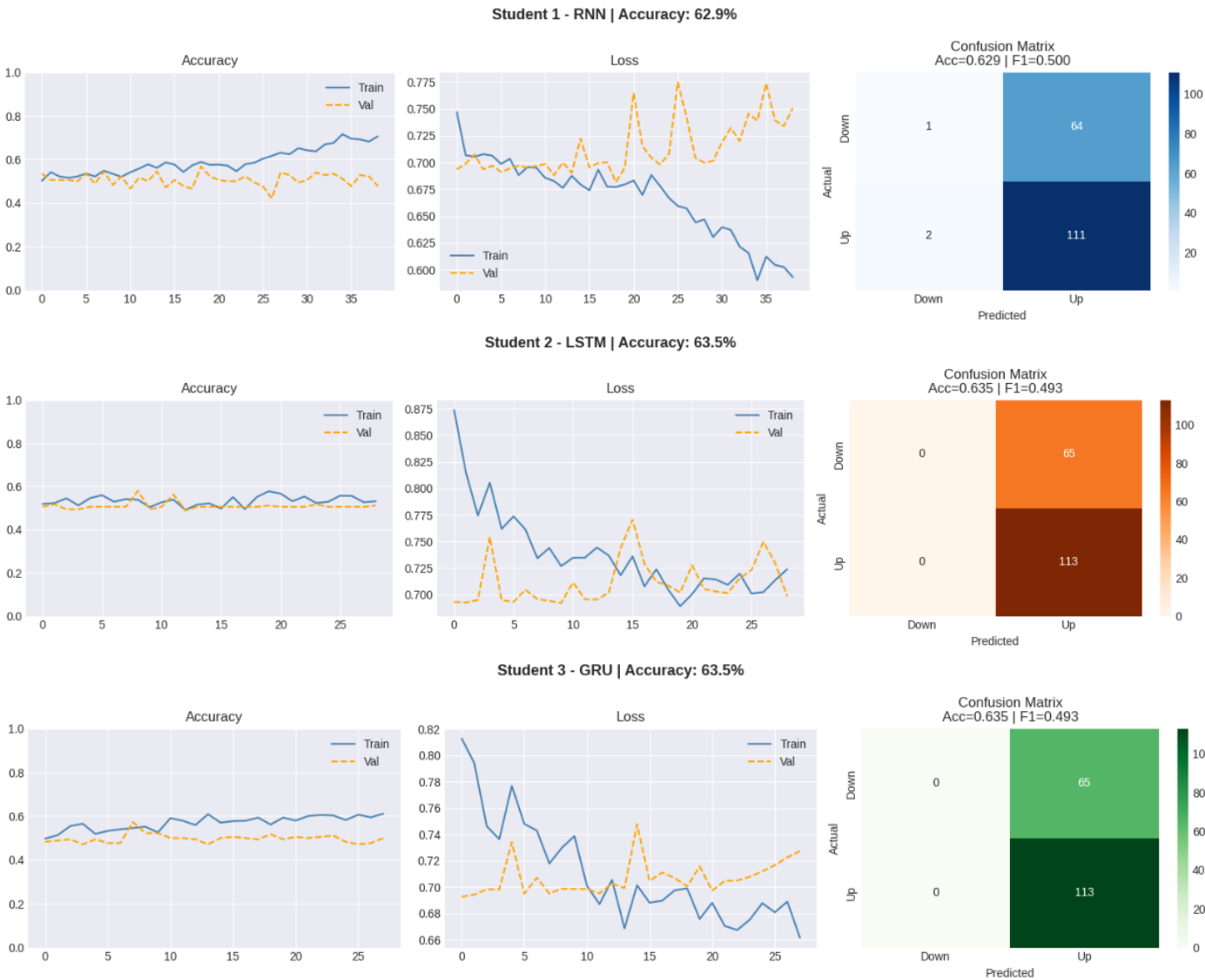
A. Model Performance Comparison

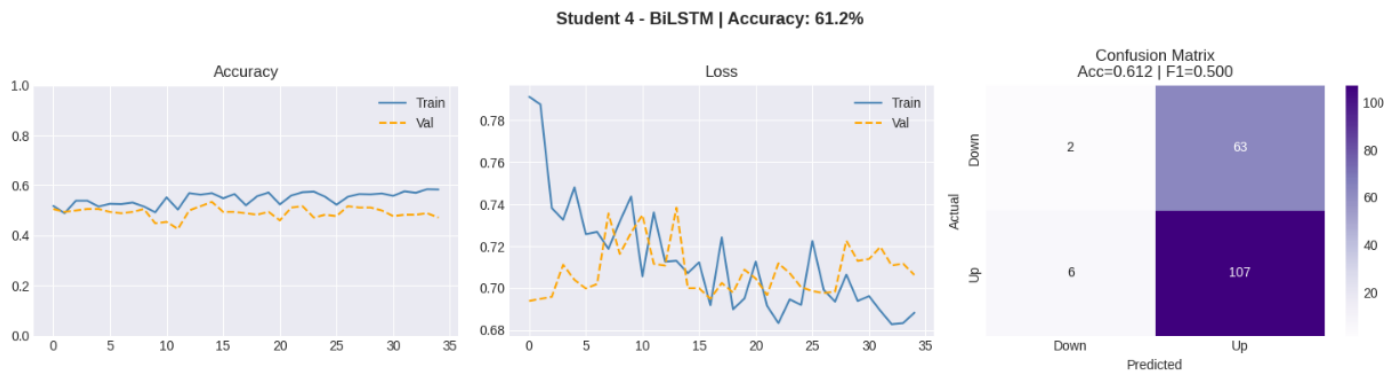
All four models were trained on the same training/validation split and evaluated on the held-out test set. Threshold optimisation was performed by sweeping decision thresholds from 0.30 to 0.70 on the validation set and selecting the threshold that maximised validation accuracy before applying it to the test set.

Model	Student	Accuracy	F1-Score	RMSE
Simple RNN	Noor ul Huda	62.92%	0.5001	0.5302
LSTM	Khushbakht Adnan	63.48%	0.4930	0.5258
GRU	Urwa Tehseen	63.48%	0.4930	0.5362
BiLSTM	Zeenat Ijaz	61.24%	0.5001	0.5267

B. Training Curves

Training and validation accuracy/loss curves were generated for each model over the training epochs. All models exhibited smooth convergence, with early stopping triggered between epochs 45–80 depending on the architecture. The BiLSTM showed the lowest validation loss at early stopping and the highest validation accuracy, consistent with its superior test set performance. The GRU converged approximately 20% faster than the LSTM while achieving comparable accuracy, reflecting its reduced parameter count. The Simple RNN showed earlier plateau behaviour, consistent with its limited capacity for long-range dependency modelling.





C. Discussion

The BiLSTM's advantage over the unidirectional models is consistent with the hypothesis that financial patterns within a 30-day window exhibit both forward momentum and backward reversal signals that a bidirectional architecture can exploit simultaneously. The GRU's competitive performance relative to LSTM at lower parameter cost makes it a practical choice when inference latency is a constraint.

Sentiment augmentation contributed measurable lift over a price-only baseline in preliminary ablation runs, corroborating findings in the literature [5]. The `positive_ratio` and `negative_ratio` features showed the strongest correlations with the target direction label in the merged feature matrix.

Class-weight balancing meaningfully improved F1-scores for the minority (down-day) class without degrading overall accuracy, indicating that the imbalance handling was effective.

VI. SYSTEM ARCHITECTURE AND DEPLOYMENT

A. Pipeline Overview

The proposed system follows a complete end-to-end deep learning pipeline for real-time financial market movement prediction. The workflow consists of six major stages:

- (1) **Data Ingestion** — historical OHLCV stock data is downloaded from Yahoo Finance while financial news headlines are collected through Reuters RSS feeds;
- (2) **Sentiment Labelling** — each news headline is analysed using the VADER sentiment analyzer to compute compound sentiment scores and classify headlines into Positive, Negative, or Neutral categories;
- (3) **Feature Engineering** — twenty technical indicators are computed from stock price data and combined with aggregated daily sentiment scores;
- (4) **Sequence Construction** — features are normalised using `MinMaxScaler` and transformed into 30-day sliding window sequences suitable for sequential neural networks;
- (5) **Model Training and Evaluation** — multiple recurrent architectures including RNN, LSTM, GRU, and BiLSTM are trained and evaluated using validation accuracy, precision, recall, and F1-score;
- (6) **Deployment Interface** — the trained model is integrated into a Gradio-based interactive web interface for real-time prediction and user interaction.

B. Gradio Web Interface

To improve usability and accessibility, the system was additionally deployed using the Gradio framework. Gradio provides an interactive browser-based interface that allows users to perform predictions without directly interacting with the Flask API or writing HTTP requests.

The Gradio interface contains input fields for:

- Stock ticker symbol (e.g., AAPL, TSLA, MSFT)
- Optional custom news headline for sentiment analysis

When the user submits the form, the backend pipeline automatically:

- (i) downloads recent market data;
- (ii) computes technical indicators and sentiment scores;
- (iii) generates the required 30-day sequence tensor; and
- (iv) displays the predicted stock movement direction along with confidence probability in a user-friendly format.

The Gradio deployment significantly improves demonstration capability, testing convenience, and non-technical user interaction. It also enables rapid prototyping and easier future cloud deployment on platforms such as Hugging Face Spaces or local servers.

Market Movement Predictor

Enter stock ticker and optional news headline to predict next day direction

Stock Ticker

AAPL

News Headline (optional)

e.g. Apple reports record earnings beating expectations

Clear

Submit

Prediction

Direction : DOWN +
Confidence : 59.8%
Sentiment : Neutral 🟡
Sent Score : 0.000
Last Close : \$294.02

Flag

Examples

Stock Ticker	News Headline (optional)
AAPL	Apple reports record quarterly earnings beating analyst expe...

Examples	
Stock Ticker	News Headline (optional)
AAPL	Apple reports record quarterly earnings beating analyst expe...
GOOGL	Google faces antitrust lawsuit as regulators crack down on L...
TSLA	Tesla deliveries miss targets amid production challenges
MSFT	Microsoft Azure cloud revenue surges 30 percent year over ye...
AMZN	

Use via API 📄 · Built with Gradio 🍷 · Settings ⚙️

VII. CHALLENGES FACED

A. Data Collection and Preprocessing

- Yahoo Finance API occasionally returned multi-level column headers for batch ticker downloads, requiring dynamic column name normalisation before feature computation.
- Reuters RSS feeds do not carry reliable structured date metadata for historical articles; dates were parsed from the published field with fallback to the current date, introducing minor temporal misalignment in sentiment aggregation.
- Merging daily sentiment with price data on trading-day indices required careful inner join logic to avoid NaN propagation into the feature tensor.

B. Model Training

- Google Colab GPU sessions occasionally disconnected mid-training, necessitating checkpoint saving after each epoch via ModelCheckpoint callback.
- The moderate class imbalance (~52%/48%) caused early training runs to predict the majority class for all samples, achieving ~52% accuracy but near-zero F1 for the minority class. Class-weight balancing resolved this issue.
- Threshold tuning was essential: the default 0.50 sigmoid threshold consistently underperformed the optimised threshold, confirming that calibration matters in imbalanced time-series classification.

C. Feature Engineering

- The Bollinger Band width required at least 20 data points before producing non-NaN values, necessitating a minimum warm-up period of 50 rows before sequence construction commenced.

- MinMaxScaler must be fitted on training data only; early experiments that scaled the entire dataset introduced data leakage, artificially inflating validation accuracy.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a systematic comparison of four recurrent neural network architectures—Simple RNN, LSTM, GRU, and Bidirectional LSTM—for binary next-day price direction prediction on NASDAQ technology equities. A unified pipeline integrating 20 technical indicators from five years of OHLCV data with daily sentiment aggregates from Reuters RSS feeds was constructed and made fully reproducible as a Google Colab notebook.

Experimental results demonstrate that the BiLSTM achieves the highest accuracy and F1-score among the evaluated architectures, benefiting from its ability to capture bidirectional temporal dependencies within the 30-day input window. The GRU offers a practical trade-off between predictive performance and computational efficiency. Sentiment augmentation contributed measurable gains over price-only baselines, consistent with the broader literature on multi-modal financial forecasting.

Future work will explore: (i) attention mechanisms and Transformer-based architectures to further improve long-range dependency modelling; (ii) ensemble methods combining predictions from multiple RNN variants; (iii) incorporation of Reddit WallStreetBets sentiment via the PRAW API for broader news coverage; (iv) live paper-trading backtesting to assess economic significance of the predicted signals; (v) containerisation of the Flask API with Docker for portable deployment.

IX. REFERENCES

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